EXPERIMENT NO. 07 - Sampling Theorem, and OFDM Modulation and Demodulation



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TITLE: Sampling Theorem, and OFDM Modulation and Demodulation.

PROBLEM STATEMENTS: :

1. **Part 1:** Consider the space of bandlimited functions, where signals are bandlimited to a maximum frequency of 30kHz. Design a sampler and a corresponding interpolator, inspired by the classical Shannon-Nyquist-Whittaken Sampling theory, and explore the performance of such a scheme (in terms of reconstruction error) for the following sampling rate choice: (i) below the Nyquist rate (undersampling or sub-Nyquist sampling), (ii) at exactly the Nyquist rate (optimal sampling or Nyquist rate sampling), (iii) above Nyquist rate (super-Nyquist sampling). Generate visual plots for the original signal (from the bandlimited class of signals), the sampled signal and the reconstructed signal.
2. **Part 2:**
   (a) Generate random bit sequences and encode using QPSK modulation
   (b) Synthesize transmit waveforms using OFDM
   (c) plot the bit error rate curves against SNR in Rayleigh fading channel using zero forcing equalizer.

BRIEF THEORY & BACKGROUND:

Sampling Theorem for Band-Limited Signals A deterministic real signal $x(t)$ with Fourier transform $X(f)$ is said to be band-limited if $X(f) = 0$ for $|f| > W$, where W is the highest frequency contained in $x(t)$. Such a signal can be perfectly represented by samples of $x(t)$ taken at a rate $f_s \geq 2W$ samples per second. The minimum sampling rate $f_N = 2W$ is called the Nyquist rate. For complex-valued signals, W represents half the total frequency support [1].

Signal Reconstruction If $x(t)$ is sampled at intervals of $1/(2W)$, then it can be exactly reconstructed using:

$$x(t) = \sum_{n=-\infty}^{\infty} x\Big(\frac{n}{2W}\Big)\, sinc\Big[2W\Big(t - \frac{n}{2W}\Big)\Big]$$

Here, $\{x(n/2W)\}$ are the signal samples taken at $t = n/2W$, and reconstruction can equivalently be achieved by passing these samples through an ideal lowpass filter with impulse response [1]:

$$h(t) = sinc(2Wt)$$

Orthogonality Property The sinc functions used in reconstruction form an orthogonal set, satisfying:

$$\int_{-\infty}^{\infty} sinc\Big[2W\Big(t - \frac{n}{2W}\Big)\Big]\, sinc\Big[2W\Big(t - \frac{m}{2W}\Big)\Big]\, dt = \begin{cases} \frac{1}{2W}, & n = m \\ 0, & n \neq m \end{cases}$$

Thus, the expansion of $x(t)$ in Eq. () is an orthogonal expansion, not an orthonormal one.

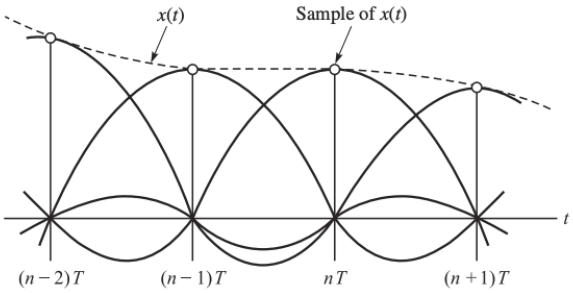


Figure 1: Sampling and reconstruction from Samples) [1].

Orthogonal Frequency Division Multiplexing (OFDM)

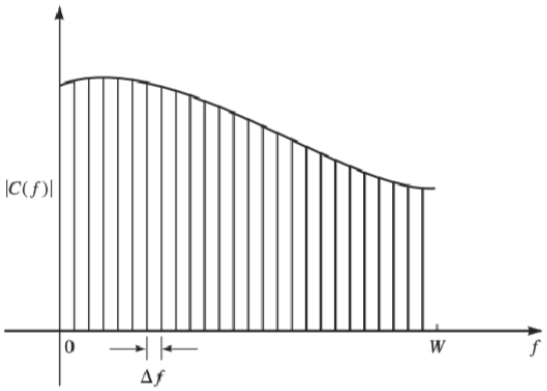


Figure 2: Subdivision of channel bandwidth) [1].

Subcarrier Structure: The available channel bandwidth W is divided into N subbands of width

$$\Delta f = \frac{W}{N}.$$

Each subband carries a sinusoidal carrier:

$$s_k(t) = \cos(2\pi f_k t), \quad k = 0, 1, \ldots, N-1$$

where f_k is the mid-frequency of the kth subchannel. The subcarriers are made orthogonal over the symbol interval T by setting

$$f_k - f_j = \frac{n}{T}, \quad n = 1, 2, \ldots, N-1$$

which ensures

$$\int_0^T \cos(2\pi f_k t + \phi_k)\cos(2\pi f_j t + \phi_j)\, dt = 0.$$

Hence, the subcarriers are mutually orthogonal regardless of the phase offsets ϕ_k and ϕ_j.
Modulation in an OFDM System: In an OFDM system with N subchannels, the symbol interval is

$$T = NT_s,$$

where T_s is the symbol duration in a single-carrier system. A larger N increases T, making intersymbol interference (ISI) negligible. Each subcarrier may be modulated using M-ary QAM. The kth subcarrier signal is

$$u_k(t) = \sqrt{\frac{2}{T}} A_{ki} \cos(2\pi f_k t) - \sqrt{\frac{2}{T}} A_{kq} \sin(2\pi f_k t) = \Re\left[\sqrt{\frac{2}{T}} X_k e^{j2\pi f_k t}\right],$$

where $X_k = A_k e^{j\theta_k}$ is the complex QAM symbol.

Channel Model: If the subchannels are narrowband, each can be modeled by a fixed complex frequency response:

$$C(f_k) = C_k = |C_k| e^{j\phi_k}.$$

The received signal on the kth subchannel is then

$$r_k(t) = \Re\left[\sqrt{\frac{2}{T}} C_k X_k e^{j2\pi f_k t}\right] + n_k(t),$$

where $n_k(t)$ is additive white Gaussian noise (AWGN).

Demodulation : At the receiver, the demodulation is performed by correlating $r_k(t)$ with two orthogonal basis functions:

$$\psi_1(t) = \sqrt{\frac{2}{T}} \cos(2\pi f_k t + \phi_k), \quad \psi_2(t) = -\sqrt{\frac{2}{T}} \sin(2\pi f_k t + \phi_k), \quad 0 \le t \le T.$$

The correlator outputs are sampled at $t = T$ to obtain

$$y_k = (|C_k| A_{ki} + \eta_{kr},\ |C_k| A_{kq} + \eta_{kq}),$$

or equivalently, in complex form:

$$Y_k = |C_k| X_k + \eta_k,$$

where $\eta_k = \eta_{kr} + j\eta_{kq}$ represents complex additive noise.

Equalization: The effect of the channel can be compensated by dividing by the magnitude of the channel gain:

$$Y'_k = \frac{Y_k}{|C_k|} = X_k + \eta'_k, \quad \text{where} \quad \eta'_k = \frac{\eta_k}{|C_k|}.$$

Thus, each subcarrier can be demodulated and detected independently, achieving orthogonal frequency-division multiplexing (OFDM).

OUTCOMES:

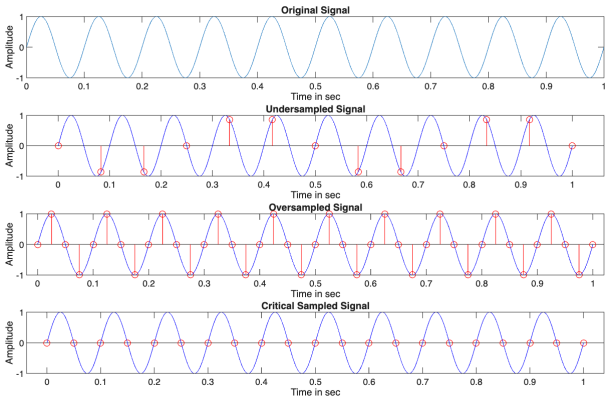


Figure 3: Original, Undersampling, Oversampling and Critically Sampled Signal.)

DISCUSSIONS:
The theory was discussed by the professor and TA. I discussed and tried to code the sampling of the signal in the lab, with my lab partner Akash. I discussed the sampling part and its reconstruction with my colleagues- Saptarshi Bhattacharya and Mayank Singh. My future work will include the reconstruction of the signal using Sampling theorem and use OFDM technique for signal transmission and reception.

ACKNOWLEDGEMENTS:
I would like to thank TA Siva Sankar for giving an introductory view on the experiment. I would like to extend my thanks to Prof Amitalok J Budkuley, for explaining the concepts, and motivating us to not rely on LLM models for code generation, to ue our logic and learn how to code. I would like to thank my lab partner Akash Kumar, with whom I was able to perform some of the prerequisites in the lab. I would like to thank my colleagues Saptarshi Bhattacharjee, Mayank Singh and Soumyajit Adak, with whom the late night sessions have turned into coding together, discussion of the concepts involved and more understanding of the material.

References

[1] J. G. Proakis and M. Salehi, *Digital Communications*, 5th. New York, NY, USA: McGraw-Hill Higher Education, 2014, ISBN: 978-0-07-295716-7.

[2] N. Anand, *Verification of sampling theorem*, `https://www.mathworks.com/matlabcentral/fileexchange/102744-verification-of-sampling-theorem`, MATLAB Central File Exchange, Retrieved November 10, 2025, 2025.

APPENDIX:
MATLAB Code [2]:
Sampling:

```
t_f = 1;
t = 0:0.00001:t_f;
f_d = 10;
% deifining signal
x_1 = sin(2*pi*f_d*t);
%plot
figure;
subplot(4,1,1);
plot(t, x_1);
title("Original Signal");
xlabel("Time in sec");
ylabel("Amplitude");
%% undersampling
fs1 = 1.2 * f_d;
%time vector
n1 = 0:1/fs1:t_f;
% generating undersampling signal
x_2 = sin(2*pi*f_d*n1);
%plot
subplot(4, 1, 2);
stem(n1, x_2 ,'r'); hold on;
plot(t, x_1, 'b'); hold off;
title("Undersampled Signal");
xlabel("Time in sec");
```

```
ylabel("Amplitude");

%% oversampling
fs2 = 4 * f_d;
%time vector
n2= 0:1/fs2:t_f;
% generating undersampling signal
x_3 = sin(2*pi*f_d*n2);
%plot
subplot(4, 1, 3);
stem(n2, x_3 ,'r');hold on;
plot(t, x_1, 'b'); hold off;
title("Oversampled Signal");
xlabel("Time in sec");
ylabel("Amplitude");

%% crticalsampling
fs3 = 2 * f_d;
%time vector
n3 = 0:1/fs3:t_f;
% generating undersampling signal
x_4 = sin(2*pi*f_d*n3);
%plot
subplot(4, 1, 4);
stem(n3, x_4 , 'r');hold on;
plot(t, x_1, 'b'); hold off;
title("Critical Sampled Signal");
xlabel("Time in sec");
ylabel("Amplitude");
```